



Ecofriendly silicon-poly(lactic acid) hybrid antifouling coatings

Fabrice Azemar*, Fabienne Faÿ, Karine Réhel, Isabelle Linossier

Laboratoire de Biotechnologie et Chimie Marines, Univ. Bretagne-Sud, EA 3884, LBCM, IUEM, F-56100, Lorient, France

ARTICLE INFO

Keywords:

Antifouling
Poly(lactic acid)
Silicon
Hybrid paints
Environmental impact

ABSTRACT

Societal and environmental constraints have led to changes in international and European regulations to limit the use of biocides in antifouling paints. In this context, coatings combining fouling release properties and leaching of active molecules are sought. The objectives of this study are to conceive a hybrid system for antifouling paint and to observe the influence of the physico-chemical properties of the binders on its antifouling activity. Poly(lactic acid) (PLA) and poly(dimethylsiloxane) (PDMS) homopolymers have already been used as binders for different antifouling strategies. The use of a block copolymer based on PLA and PDMS is likely to allow the design of paint with biocides release associated to fouling release properties in order to be more efficient with a lower environmental impact.

Triblock copolymers with different molar compositions have been synthesized. The properties of the corresponding films have been determined in distilled water (adhesion to the support, hydrophobicity). Then formulated paints have been immersed in Lorient harbor to evaluate their erosion, degradation and antifouling efficacy. Furthermore, their biocide release kinetics were studied in artificial seawater to evaluate their environmental impact.

The silicon-poly(lactic acid) paints have shown an efficiency superior to a commercial paint during their *in situ* immersion (19 months) with a lower biocides amount (14 %).

1. Introduction

Since the 19th century and the use of first antifouling paints [1], lots of researches have been done to increase their efficiency against the biofouling and their lifetime [2,3]. However, these improvements have been done to the detriment of the environment. The more known example is the introduction of tributyltin-based compounds (TBT) as biocide in paints formulation to obtain cheap paints with an efficiency of 5 years [4]. Unfortunately, their use has caused many troubles on marine species due to the accumulation and severe toxicity of TBT [5,6]. Currently, new alternatives are based on metals compounds such as copper or zinc [7]. The toxicity of these biocides has been widely evaluated [8–10]. Mochida et al. [8] have shown that copper is a poison for algae and mollusks. Nevertheless, these antifouling alternatives continue to release a large amount of biocides into seawater [11]. The bioaccumulation into seawater environment generated by the leaching has been measured around the world [12–14] and has been shown as the cause of some environmental problem or genetic modification [15]. In the future, regulations and legislations should prohibit some biocides and limit the use of the others to decrease the impact of chemicals on the environment.

The fouling release strategy based on paints with physico-chemical

surface properties to limit durable colonization by organisms is a promising alternative [16]. Unfortunately, these paints are inefficient during static periods [17,18]. Indeed, the biofilm is removed by hydrodynamic stress like navigation for ships or a mechanical cleaning for static structures. Oil addition is a common technique for enhancing the biofouling resistance properties of fouling release coating [19]. Silicone oils incorporated into the coating reduced the adhesion strength of marine species [20]. As biocides, silicone oils are hazardous substances and a potential threat to seawater environment [21].

In the aim to develop an efficient and more ecological antifouling paint, our group have recently developed hybrid antifouling coatings [22]. These paints were able to control the release of biocides due to the degradation properties of poly(ϵ -caprolactone) (PCL) [23,24], but they had also surface properties brought by the poly(dimethylsiloxane) (PDMS). PDMS is the most found silicone in commercial application and it has done good results as binder in antifouling paints with surface properties [25,26]. Hybrid paints were able to prevent the fouling proliferation with a low amount of biocides. Nevertheless, the high crystallinity of PCL, a key parameter, was liable for the short life of paint.

In the present work, to increase the paint life, linear copolymers contained poly(D,L-lactic acid) (PLA) and PDMS have been synthesized.

* Corresponding author.

E-mail address: fabrice.azemar@univ-ubs.fr (F. Azemar).

<https://doi.org/10.1016/j.porgcoat.2020.105841>

Received 19 March 2020; Received in revised form 15 June 2020; Accepted 15 June 2020

Available online 20 June 2020

0300-9440/© 2020 Elsevier B.V. All rights reserved.

PLA is amorphous and well known for its high degradability [27,28]. Moreover, lots of researches have been done to use this polyester in antifouling strategy [29–31]. The copolymers with external PLA blocks and PDMS core block were incorporated as binders in antifouling coating to evaluate their antifouling efficiency in seawater. The influence of the block size on the coatings properties was measured such as the degradation and the surface properties. Furthermore, a study of the metals leaching during the immersion has been done to show the low environmental impact of hybrid paints compared to commercial paints.

2. Experimental

2.1. Materials

The lactide monomer (LA) was acquired from TCI (Tokyo, Japan). The catalyst stannous octanoate ($\text{Sn}(\text{Oct})_2$) was purchased from Sigma-Aldrich (Saint-Louis, MO, USA). The polydimethylsiloxane Tegomer® H-Si 2311 ($M_n \approx 2300 \text{ g.mol}^{-1}$) was kindly supplied by Evonik Industries, Germany. The marine salt (Crystal Reef) used to prepare the artificial sea water (ASW) was purchased from Philanima (France). All the products were used as received.

2.2. Copolymers synthesis and characterization

Copolymers are triblock copolymers with poly(DL-lactic acid) external blocks and a polydimethylsiloxane core block. The copolymers were synthesized by ring opening polymerization of LA in the presence of PDMS with $\text{Sn}(\text{Oct})_2$ as catalyst (Fig. 1). The copolymer composition was varied by changing the feed ratio of the reactants [LA]/[PDMS] (304:1; 144:1; 64:1; 37:1; 24:1; 16:1 n/n respectively) to study the effect of the composition on the antifouling properties (Table 2). The PDMS molecular weight was set as 2300 g.mol^{-1} . Only the size of the PLA block changed. The same procedure as described in a precedent work was used [32]. Copolymers synthesis occurred at 130°C under nitrogen during 12 h and the ratio of $[\text{Sn}(\text{Oct})_2]/[\text{PDMS}]$ used is 2:1 n/n.

2.2.1. Gel permeation chromatography (GPC)

GPC chromatograms were measured by an Agilent Technologies 1200 Infinity II chromatography containing an isocratic pump, a column oven at 35°C and a RID detector equipped with a Varian 410 autosampler. The separation was performed by two columns PLgel (Mixed-E, $3 \mu\text{m}$ and Mixed-D, $5 \mu\text{m}$) from Polymer Laboratories and a column guard. THF was used as eluent at a flow rate of 0.8 mL.min^{-1} , and the injection volume was $20 \mu\text{L}$. The calibration was done with polystyrene standards with molar masses included between 1300 and $400,000 \text{ g.mol}^{-1}$ (Easical PS-2) from Agilent Technologies. To measure the copolymer degradation into the paints, the dried paints were diluted with THF. The mixtures were centrifuged at 5000 rpm during 10 min at 20°C to eliminate the paint fillers and additives. Then the supernatants were filtered with PTFE filter ($0.45 \mu\text{m}$)

2.2.2. Nuclear magnetic resonance of proton (^1H NMR)

^1H NMR spectra of the copolymers were obtained on a Bruker Avance 500 spectrometer at 500 MHz. Spectra were taken in deuterated chloroform at 30°C . ^1H NMR (ppm): 0.06 [$\text{Si}(\text{CH}_3)_2$, 6 H], 1.5 [$\text{COCH}(\text{CH}_3)\text{O}$, 3 H], 5.1 [$\text{COCH}(\text{CH}_3)\text{O}$, 1 H]. The composition of copolymers

Table 1
Components of the formulation.

Solvents	Xylene MIBK	35 %
Binder	Copolymers	20 %
Fillers	TiO_2 ZnO CaCO_3	30 %
Additives	Zinc pyrithione Copper thiocyanate Polyamide waxes	2 % 12 % 1 %

was calculated from the ratios of intensities at 0.06 ppm and 5.1 ppm.

2.3. Paint preparation

To evaluate the physico-chemical and antifouling properties of the binders, paints were formulated with the six copolymers and with an homopolymer of PLA used to evaluate the PDMS influence. The formulation used was adjusted from the formulation of the commercial paint used as reference for the evaluation of antifouling efficiency. All the components presented in the Table 1 were dispersed under 600 rpm during 30 min. Copper thiocyanate (CuSCN) and zinc pyrithione (ZnPT) were used as biocide and polyamide waxes was introduced to avoid the sedimentation. Then the paints were applied on polycarbonate plates with an automatic applicator (ASTM D823 – $500 \mu\text{m}$) for the studies of the physico-chemical properties and with a brush on polyvinylchloride plates for the evaluation of antifouling properties in seawater ($12 \times 14 \text{ cm}$).

2.3.1. Scanning electron microscopy (SEM)

To observe the paint surface, the paints were observed with an electron microscope JEOL 6460 LV equipped with an Oxford Inca 300 X-Ray microanalysis. To study the surface, the dry paint was directly metalized with carbon. For all investigations, the beam energy was 18 kV in order to obtain the excitation of the element.

2.3.2. Contact angle measurement

Captive bubble method was used to evaluate the hydrophilic properties. Measurements were taken at room temperature with a contact angle meter Digidrop GBX. Five air bubbles of $3 \mu\text{L}$ were dropped at different sites on each paint, and the measured values of the contact angle were averaged. A camera was used to observe the air bubble profile.

2.4. Copper leaching

To evaluate the copper release into water by paints containing PLA-PDMS copolymers, a paint layers of $8 \times 22 \text{ cm}$ were applied on polycarbonate support with the automatic film applicator. Then the coatings were fixed on a polycarbonate cylinder ($\Phi 64 \text{ mm}$ and L of $170 \pm 2 \text{ mm}$). Each paint was immersed in artificial seawater in its own 21 beaker. At each sampling dates, the cylinder was fixed on a rotating device. Rotating device was able to rotate the test cylinders in the measuring container at 60 rpm during 1 h. The device itself was not

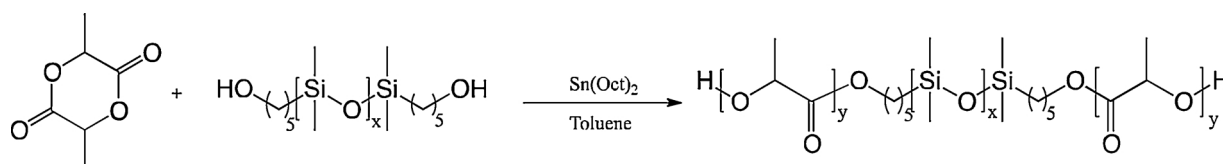


Fig. 1. Synthesis of the binder by ring opening polymerization of lactide.

Table 2
Characterization of copolymers.

Copolymers	Theoretical molar ratio ^a PLA- <i>b</i> -PDMS- <i>b</i> -PLA (%)	Molar ratio of PLA calculate by ¹ H NMR ^b (%)	Theoretical molar masses (g mol ⁻¹)	Molar masses by ¹ H NMR (g mol ⁻¹)	Molar masses by GPC (g mol ⁻¹) ^c	$\bar{M}_n$
C 55	27- <i>b</i> -45- <i>b</i> -27	57	5000	5300	6500	1,26
C 66	33- <i>b</i> -33- <i>b</i> -33	66	6800	6760	8500	1,44
C 75	37- <i>b</i> -25- <i>b</i> -37	74	8950	8890	12500	1,39
C 80	40- <i>b</i> -20- <i>b</i> -40	82	11200	13130	17500	1,36
C 90	45- <i>b</i> -10- <i>b</i> -45	91	22300	25880	26500	1,44
C 95	47- <i>b</i> -5- <i>b</i> -47	96	44500	50950	40000	1,5
H 10000	—	100	10000	9500	11000	1,5

^a Calculated with the starting molecular ratio of LA and PDMS.

^b Calculated with the relative intensities of the peak at 0.06 ppm and at 5.1 ppm.

^c In THF as eluent and with polystyrene standard.

into contact with the artificial seawater to avoid pollution.

The cumulative method was used to measure the copper leaching. A leaching water sample was analyzed before and after dynamic phase. Artificial sea water was changed at each sampling dates in order to prevent the risk of medium saturation.

2.4.1. Voltammetry measurement

Copper concentrations released by paints during the leaching were measured using a Metrohm Voltammetry Analyser 884 Professional VA equipped with a 919 IC Autosampler. All analysis were carried out using a standard solution of Cu at 2 mg.L⁻¹ and an electrolyte solution of KCl with a pH of 4.6. The standard addition method and the hanging mercury drop electrode (HDME) as working electrode were chosen to perform measurement.

3. Results and discussion

3.1. Copolymers synthesis and characterization

As in our previous work [32], copolymers were synthesized by ring-opening polymerization of D,L-lactide to obtain a paint with surface and release properties. PDMS macroinitiator with two hydroxy end-groups allowed the synthesis of triblock copolymers with PLA end-blocks and a PDMS core-block (Fig. 2). The biodegradable properties of external PLA block gave erodible properties to the paints to control the release biocides. PDMS block gave surface properties to the coating to get fouling release ability. Six copolymers were prepared to study the influence of the molar ratio on antifouling efficiency and paints properties. For example, C 55 means that the theoretical ratio of PLA in the copolymer is 55 %. The syntheses were carried out at 130 °C for 12 h to obtain a complete conversion of the monomer. Copolymers synthesized were analyzed by GPC, ¹H NMR and FTIR to evaluate the structure and the molar ratio of PLA and PDMS. Monomodal peaks observed on GPC chromatograms (Fig. 3) and molar masses measured by GPC (Table 2)

confirmed the copolymers synthesis. Moreover, the absence of lactide protons peaks on the ¹H NMR spectra showed that all lactide monomers have reacted. As shown in Table 2, ¹H NMR was used to calculate the PLA molar ratio for each copolymer and the molar masses. GPC allowed the determination of molar masses and polydispersity. The molar ratio of PLA and molar masses calculated confirmed that all copolymers were synthesized with the expected structure. The difference between molar masses obtained by GPC and ¹H NMR derived from the use of polystyrene standard for GPC analysis. ¹H NMR spectra of purified copolymers had no peak of octanoate protons. It seemed that all tin was removed.

3.2. Coatings characterization

The major properties of the paints came from the binder [33]. Here, the triblock copolymers were mainly responsible of the hybrid properties. Nevertheless, the interactions between all the paints components could have an impact on the stability of a formulation and its film-forming properties. The choice of the fillers and additives introduced in a formulation depends upon the paints purpose. It will be different for a biocide release paint and a fouling release coating. The formulation used for this project promoted the erosion and has been already used for polyester antifouling [34]. To evaluate the formulation used, the adhesion of the coating on the PVC substrate in seawater and the integrity of the paint surface were observed. Moreover, the PDMS influence on the surface properties was quantified to confirm the hybrid properties of the coatings.

3.2.1. Film-forming properties

Two difficulties were likely to be encountered due to the presence of PDMS in the polymers. First, a problem of adhesion to the support. PDMS-based coatings, due to their surface properties, require a specific primer coat for good adhesion. Second, PDMS are used in a cross-linked form to obtain satisfactory mechanical properties. In the context of this

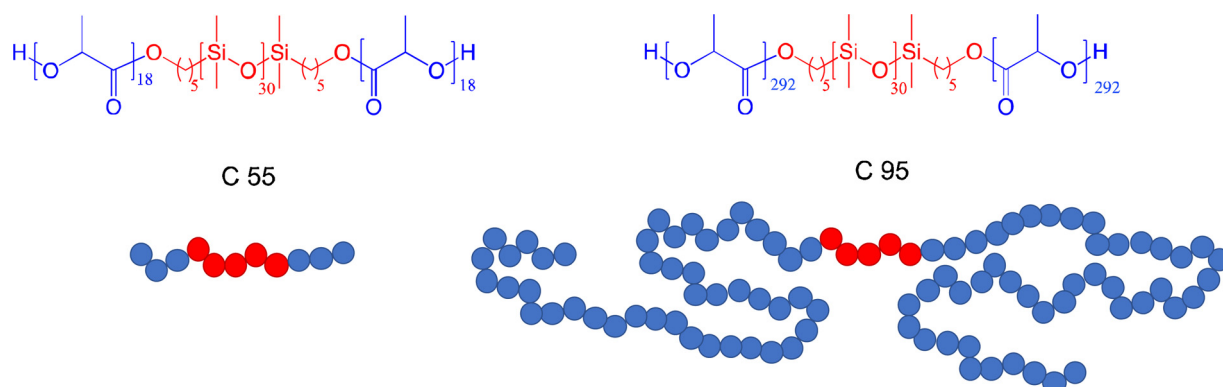


Fig. 2. Copolymers structure and influence of the molar ratio on the PLA block length.

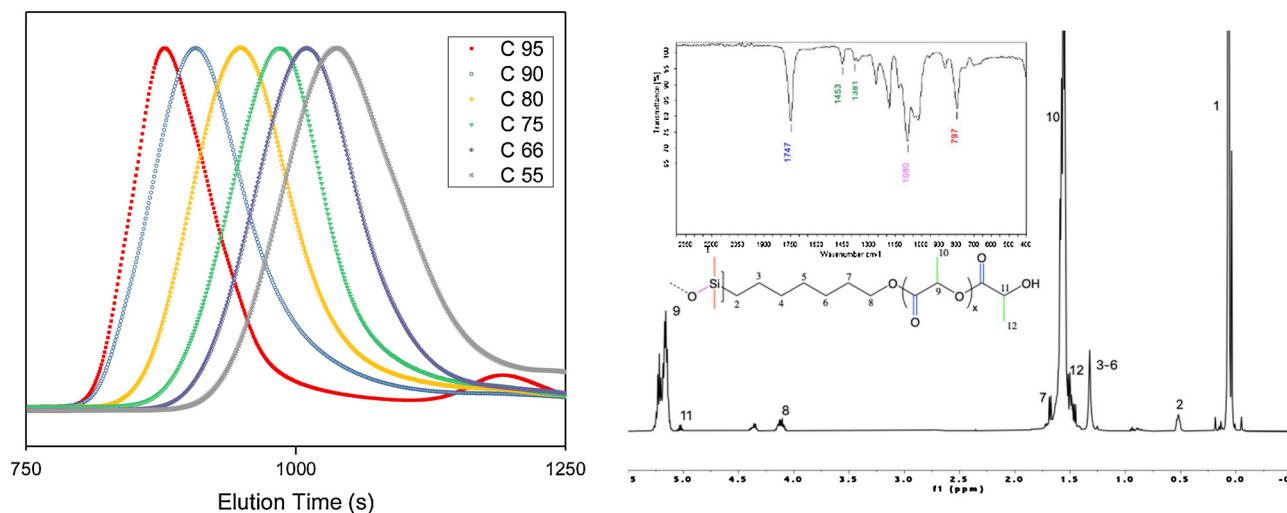


Fig. 3. Copolymers chromatograms, FTIR and NMR ^1H spectrum after drying.

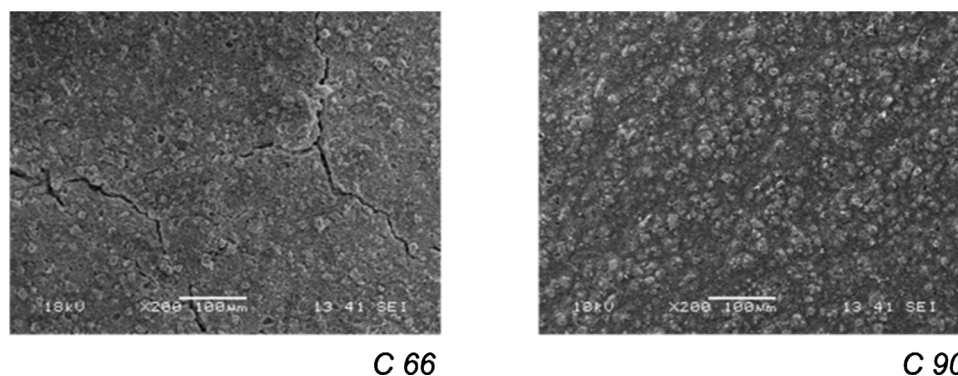


Fig. 4. Antifouling paints surfaces observed by SEM.

work, the copolymers were used without cross-linking and without prior coating. Pictures obtained by scanning electron microscopy showed cracks on the paints after one week in seawater for the copolymer C 66 compared to the homogeneous surface of paint C 90 (Fig. 4). These cracks could promote the water absorption leading to a reduction of film cohesion and a shortening of the paint lifetime. The similar surface with cracks was observed for the paint using the copolymer C 55 as binder. Robust coatings were obtained when the molar ratio of PLA is superior to 75 %. Moreover, the same trends were observed concerning the adhesion of paints. During the first month of immersion in seawater, the paints containing the copolymers C55 and C 66 showed a loss of adhesion on the PVC substrate. The other paints remained unaltered. The results underline that the molar ratio of PDMS and certainly the molar mass of the copolymer were keys parameters to obtain good film-forming properties.

3.2.2. Surface properties

In order to ensure that analysis conditions were as close as possible to the actual conditions of use, the captive bubble method was used to assess the surface properties of the coatings. This method allows the measurement of the contact angle of an air bubble with the immersed surfaces. Moreover, it gives reproducible contact angles which are essentially independent of measurement times and drop volumes [35]. Paints were immersed in distilled water during 9 months to observe their hydrophilicity evolution.

Fig. 5 presents the contact angle obtained after the first day of immersion. Two reference surfaces have been used: the first, a hydrophilic one is glass; the second, a hydrophobic one, is PDMS. Average contact angles of the air bubble of 143° and 110° were measured respectively.

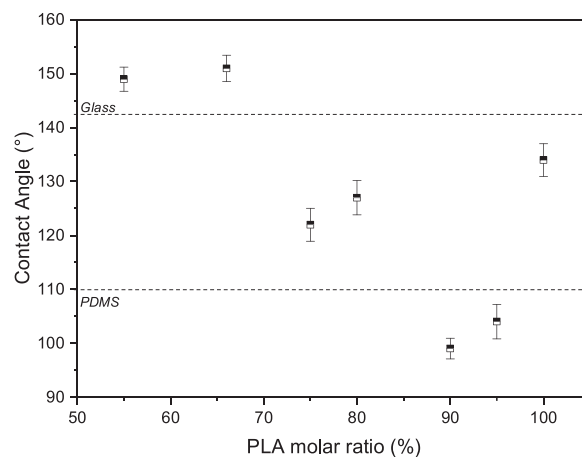


Fig. 5. Contact angle values by captive air bubble method in distilled water on the paints surface.

The coatings containing the copolymers with the biggest ratios of PDMS, C55 and C66, had the highest contact angles (152°). These values were probably due to the presence of water in the paint. The cracks observed by SEM on their surfaces eased the water absorption and diffusion. The other paints had a surface more hydrophobic than the PLA one (134°). A contact angle inferior to the PDMS one was even obtained for C90 and C95 with respectively 99° and 103° . This result should be explained by the combination of two phenomena. On the one hand, the incompatibility of the PLA and PDMS chains which leads to

the formation of micro-domains; and secondly, the influence of the size of PLA chains on the re-organization of the segments and the appearance of crystallin domains [36]. The hydrophobicity of C90 and C95 was due to the conjugation of the hydrophobicity of PDMS and the crystallinity of the PLA micro-domains. Indeed, a big ratio of PDMS did not give the expected surface properties and seemed problematic for the formulation used.

During the next weeks of immersion (data not shown), the contact angles measured for each hybrid paint showed an increase of hydrophilicity until a stable contact angle of 150°. This evolution was caused by the water absorption and the presence of water at the surface. As a reminder, the water absorption is one of the first event occurring after immersion of coating and the hydration has an influence on the paint erosion and biocides release [37]. The paints containing the copolymers with a low ratio of PDMS needed more time to reach a contact angle of 150°. For example, the paint C 75 and the paint C 95 needed respectively one and eight months. The copolymer C 95 was the binder with the best properties, it allowed the obtention of paint with a hydrophobic surface close to the PDMS and a low water absorption.

To conclude, the composition of the binder (PDMS ration and length of PLA chains) seemed to have an influence on the coating. In this condition, to obtain an efficient coating with a good lifetime, the molar ratio of PDMS in the copolymer had to be superior to 75 %. It is coherent with the physical state of the copolymers obtained after purification. C 55 and C 66 were viscous while the others were solid. The necessary crosslinking step already used for the PDMS in fouling release system should be a solution to increase the behavior of these two coatings.

3.3. Paints properties during immersion in seawater

In theory, PDMS should have impacted the properties of the coating such as hydrophobicity and the elastic modulus, which determine the adhesion strength of organisms and the fouling release ability. Furthermore, PLA, due to its biodegradability, should have brought erosion and the controlled release of biocides. The contribution of PDMS to hydrophobicity has already been demonstrated in the previous paragraph. Its impact on cleaning ability was assessed during *in situ* immersion tests. The contributions of PLA were assessed by studies of paint erosion, polymer degradation and the kinetics of release of a copper-based biocide. The copolymers were added under a vigorous agitation with all components of the Table 1 and applied on PVC plates to obtain antifouling coatings. Their antifouling efficiency was measured by immersion in static condition in the Atlantic Ocean (Lorient Harbour) during 19 months.

3.3.1. Erosion

Erosion promotes the biocide release and the surface renewal. During the *in situ* immersion, a touching method was used to evaluate the erosion ability. The low adhesion of paints C 55 and C 66 did not allow to determine the erosion. For the other hybrid paints, the beginning of coating erosion was observed between the fifth and the sixth month. The composition of the copolymers seemed to have a low influence on the erosion beginning. The formulation was more relevant. The soluble fillers and additives containing in the paint were dissolved with the water diffusion throughout the coating. Nevertheless, shorter the PLA blocks were, the more paint was removed by touching. The lifetime was better for the paints containing the longest PLA blocks.

3.3.2. Copolymer degradation

The paint degradation comes mainly from the copolymer degradation. Moreover, self-polishing antifouling paints use degradable binder to promote the biocide release [2]. Polyester-*b*-PDMS-*b*-polyester triblock copolymers degradation occurs by the end of polyester block in small oligomers [32] and two ways of degradation are known for the PLA; a degradation by the hydrolysis of esters moieties and an

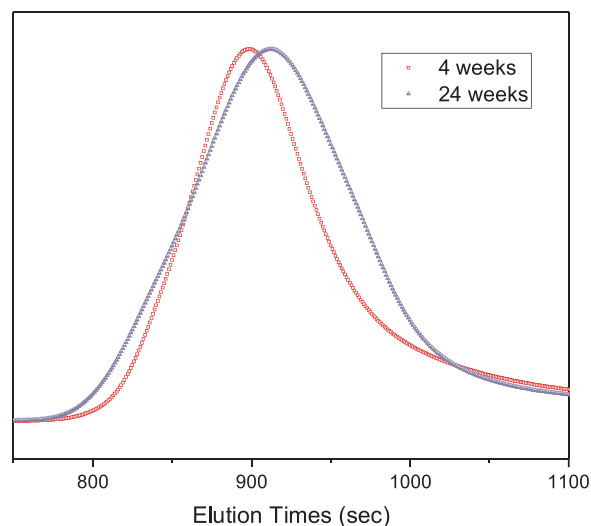


Fig. 6. Degradation of copolymer C90 in paint during immersion in seawater observed by gel permeation chromatography.

enzymatic degradation. To measure the degradation of the copolymers in the paint, they were extracted from the paint and analyzed by GPC.

GPC chromatograms of the copolymer C 90 in the Fig. 6 were obtained after 4 weeks and 24 weeks in seawater. Two differences on the chromatograms confirmed the degradation of the copolymer. The increase of elution time after 24 weeks was due to the mass loss. And the increase of the peak width was due to the increase of polydispersity. The same results were observed from the chromatograms of other copolymers (data not shown). All copolymers contained in the paints showed a degradation of PLA blocks.

The study of the copolymers degradation during the first month of immersion (Table 3) was coherent with the erosion and the contact angle measurements. The more hydrophobic coatings (C 90 and C 95) were not degraded after one month. During this period, it seemed that the degradation was mainly a hydrolysis of ester moieties. The polyester hydrolysis needed water and the coating C 90 and C 95 have shown a low kinetics of water absorption during the captive bubble measurements. Reversely, the more hydrophilic the paint was, the faster the kinetics of water absorption was, and the more the copolymer was degraded. Moreover, these results showed the interest to synthesize and use triblock copolymers with a PDMS core block. The paint containing the PLA homopolymer had lost 11 % of molecular weight while the paint C 66 had lost 4,5 % and while the degradation of the copolymer C 90 had not begun.

3.3.3. Biocides leaching

Biocides allow an antifouling efficiency but they have a detrimental environmental impact. Their release needs to be controlled to decrease this impact. Two biocides were introduced in the hybrid paints, a booster biocide (ZnPT) and a copper derivative (CuSCN). The choice to measure only the copper release in the leaching water was due to the low quantity of zinc ion introduced in paints (1 %) and to the ubiquity of zinc in the formulation.

The concentrations of copper measured in the surrounding water

Table 3

Evolution of the binder molecular weight during the first month of *in situ* immersion in seawater.

Paint	66	75	80	90	95	PLA
Shift of peak molecular weight in one month (g mol ⁻¹)	500	400	350	0	0	2050
Molecular weight lost in one month (%)	4.5	2.5	1.5	0	0	11

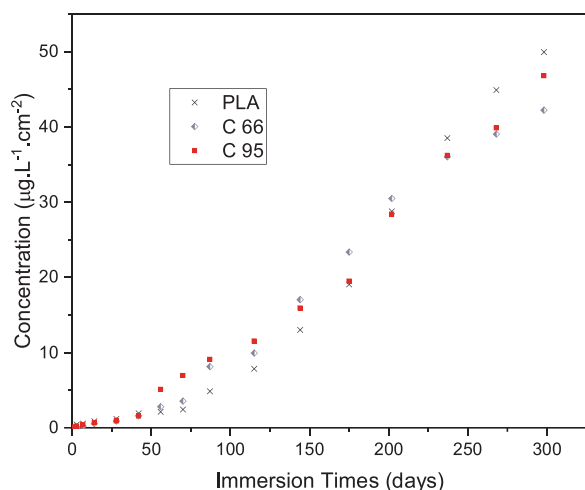


Fig. 7. Cumulative copper concentration release by paints during the leaching in artificial seawater.

showed a good control of the biocide release (Fig. 7). This behavior seemed to come from the PLA properties. Hence, the hybrid paints containing the copolymers had the same kinetics of release as the paint containing the PLA homopolymer. The copper release was practically constant during the immersion. After 300 days in artificial seawater, the cumulative concentrations of copper released by each paint were close (between 40 and 50 $\mu\text{g.L}^{-1}.\text{cm}^{-2}$). Nevertheless, the paints with the lower ratio of PLA released less copper. To compare with previous studies done in the laboratory on commercial antifouling paints, hybrid paints had a lower environmental impact. Using the standard test for paints and varnishes [38,39], commercial paints have shown a cumulative concentration of copper between 250 and 500 $\mu\text{g.L}^{-1}.\text{cm}^{-2}$ in 45 days. During the same period, hybrid paints have released approximately 5 $\mu\text{g.L}^{-1}.\text{cm}^{-2}$ of copper. Moreover, the standard test was less drastic than ours. Indeed, in this protocol, measurements were done only after a short dynamic phase and, mostly, before each dynamic phase the paint was cleaned. Adeleye et al. [40] have studied the leaching of a commercial antifouling paint using 47,6 % of copper oxide as biocide. Their results have shown releases in seawater superior to 7 $\text{mg.L}^{-1}.\text{cm}^{-2}$ after 180 days using a cumulative method. A major difference between the hybrid paints developed in this project and the commercial ones was the rate of biocides introduced. Commercial antifouling paints have a rate of biocides between 25 and 80 % [40,41]. This rate was superior to the biocide amount introduced in the hybrid paints (14 %). The lower rate combined with the properties of the paint seemed to be responsible for the low biocide release and the lower environmental impact.

3.3.4. Antifouling efficiency

To evaluate antifouling activity, the paints were immersed in static condition in the Atlantic Ocean (Lorient Harbour) during 19 months. The immersion started in March 2017 to have an observation during two summers which are the season with the most fouling development. The *in situ* conditions provided a more complete evaluation than lab-test based on specific organisms cultivated in controlled conditions. The site was previously calibrated; the seasonal fluctuations of seawater parameters (pH, temperatures and biofilm communities [42]) were measured. In the present work, only one step of the development process of marine fouling described by Haras [43] was observed; the proliferation of the macrofouling. According to a published study about antifouling evaluation, paints were evaluated by comparison with an efficient commercial paint based on the same biocides and with a PVC support without any paint [44]. The paint reference used was a controlled erodible coating with copper thiocyanate (CuSCN) and zinc pyrrithione (ZnPT) as biocides. The total percentage of copper and zinc

in the paint reference was superior to 30 %. In comparison the percentages of copper and zinc were respectively 12 % and 2 % in the hybrid paints.

The visual inspection of paints during immersion in seawater allowed the evaluation of antifouling efficiency against the macrofouling. Pictures obtained (Fig. 8) showed antifouling properties of hybrid paints during 19 months. Quickly the PVC substrate without paint was completely covered by macro-organisms. This result validated the relevance of the selected site of immersion. Both paints C 55 and C 66 with a poor adhesion have a lifetime of only three months. The comparison between other paints containing PLA-PDMS copolymers and the commercial reference showed that their antifouling activity could be higher depending on the PLA molar ratio. After 15 months in seawater, hybrid paints C80, C90 and C95 had a surface with a low colonization. Nevertheless, the paint lifetime due to the erosion was different. For example, after 19 months, paint C 95 still had a lower colonization compared to the commercial reference while all the other paints containing hybrid copolymers were completely eroded. There was no more film on the PVC plates. These results were coherent with the degradation of the copolymers measured previously and the PLA block size.

To evaluate the impact of PDMS in the coating, a paint using the same formulation and a homopolymer of PLA (10 kg.mol^{-1}) was immersed. Pictures of Fig. 8 showed good antifouling efficiency for this paint. The behavior was better than the mostly hybrid paints. After 8 months, only two hybrid paints had a lower fouling (C90 and C 95). The PDMS influence was mainly observed at the eighth month. The PLA paint showed a cleaning with the use of the water jet, but pieces of paint were removed. A low hydrodynamic stress degraded the paint while only fouling was removed on hybrid paints (C95). It seemed that PDMS provided the desired surface properties.

4. Conclusion

The interest to develop hybrid coating was to decrease the environmental impact with a good paint lifetime and a good antifouling efficiency in static and dynamic conditions. The use of a copolymer with poly(D,L-lactic acid) and poly(dimethylsiloxane) as binder has led to a paint that associates surface interest and biodegradable properties. The corresponding paints were able to prevent fouling settlement and proliferation by i) erosion, ii) biocides release and iii) low hydration kinetic due to a hydrophobic surface. This study has shown that the molar ratio of PLA and the length of the PLA block were key parameters. The best results were obtained for the paint containing a low molar ratio of PDMS and the longest PLA block. The PDMS core block was essential, it had an influence on the hydrophobicity and the degradation of PLA block. The decrease of environmental impact was a major aim of this study. The significantly decrease of biocide release observed during immersion confirmed the choice to develop hybrid paints.

CRedit authorship contribution statement

Fabrice Azemar: Conceptualization, Methodology, Investigation, Resources, Writing - review & editing. **Fabienne Faÿ:** Supervision, Validation. **Karine Réhel:** Funding acquisition, Writing - review & editing. **Isabelle Linossier:** Funding acquisition, Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

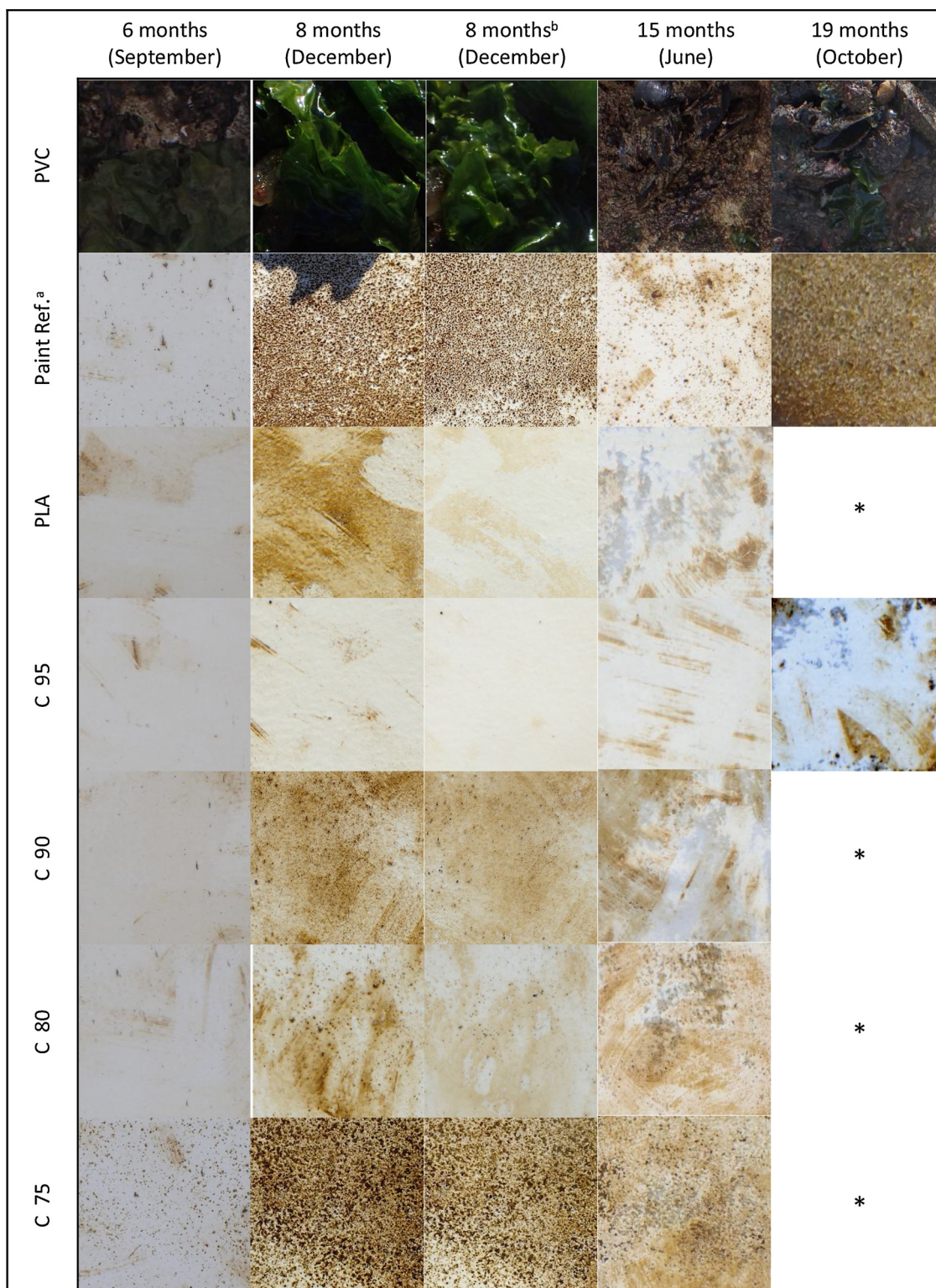


Fig. 8. Macrofouling development observed on the paints surface (6 × 6 cm) during the *in situ* immersion from March.

^a Commercial paint with biocides.

^b Pictures were taken after cleaning with a low water jet.

* No paint on the PVC plate.

Acknowledgements

This work was supported by ANR SAFER 15-LCV4-0006. We sincerely thank Evonik Tego Chemie GmbH Company for the kind gift of the functional silicone monomer (Tegomer® H-Si 2311). This

manuscript is a tribute to the 50 year anniversary of the French Polymer Group (Groupe Français des Polymères - GFP).

Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.porgcoat.2020.105841>.

References

- [1] S. De Nancay, Nav. Ports Chantiers 39 (1989) 238.
- [2] M. Lejars, A. Margaillan, C. Bressy, Fouling release coatings: a nontoxic alternative to biocidal antifouling coatings, Chem. Rev. 112 (2012) 4347–4390.
- [3] X. Yang, L. Yan, F. Ran, Y. Huang, D. Pan, Y. Bai, L. Shao, Mussel-/diatom-inspired silicified membrane for high-efficiency water remediation, J. Membr. Sci. 597 (2020) 117753.
- [4] K.A. Dafforn, J.A. Lewis, E.L. Johnston, Antifouling strategies: history and regulation, ecological impacts and mitigation, Mar. Pollut. Bull. 62 (2011) 453–465.
- [5] IMO, International Convention on the Control of Harmful Antifouling Systems on Ships, International Maritime Organisation, London, 2001.
- [6] C.L. Alzieu, J. Sanjuan, J.P. Deltreil, M. Borel, Tin contamination in Arcachon bay: effect on oyster shell anomalies, Mar. Pollut. Bull. 17 (1986) 494–498.
- [7] I. Amara, W. Miled, R. Ben Slama, N. Ladhari, Antifouling processes and toxicity effects of antifouling paints on marine environment. A review, Environ. Toxicol. Pharmacol. 57 (2018) 115–130.
- [8] K. Mochida, K. Ito, H. Harino, A. Kakuno, K. Fuji, Acute toxicity of pyrrhione antifouling biocides and joint toxicity with copper to red sea bream (*Pagrus major*) and toy shrimp (*Heptacarpus futilirostris*), Environ. Toxicol. Chem. 25 (2006) 3058–3064.
- [9] V.W.W. Bao, K.M.Y. Leung, J.W. Qiu, M.H.W. Lam, Acute toxicities of five commonly used antifouling booster biocides to selected subtropical and cosmopolitan marine species, Mar. Pollut. Bull. 62 (2011) 1147–1151.
- [10] P.A. Turley, R.J. Fenn, J.C. Ritter, M.E. Callow, Pyrrhiones as antifoulants: environmental fate and loss of toxicity, Biofouling 21 (2005) 31–40.
- [11] A.S. Adeleye, E.A. Oranu, M. Tao, A.A. Keller, Release and detection of nanosized copper from a commercial antifouling paint, Water Res. 102 (2016) 374–382.
- [12] B. Misson, C. Garnier, B. Lauga, D.C. Dang, J.F. Ghiglione, J.U. Mullot, R. Duran, O. Pringault, Chemical multi-contamination drives benthic prokaryotic diversity in the anthropized Toulon Bay, Sci. Total Environ. 556 (2016) 319–329.
- [13] S. Lee, Y.W. Lee, Determination of the concentration of alternative antifouling agents on the Korean coast, Mar. Pollut. Bull. 113 (2016) 253–257.
- [14] M.N.R. Rosli, S.B. Samat, M.S. Yasir, Analysis of heavy metal accumulation in fish from the coastal waters of Terengganu, Malaysia, AIP Conf. Proc. 1940 (2018) 020010.
- [15] J. McBain, A.H. Rickard, P. Gilbert, Possible implications of biocide accumulation in the environment on the prevalence of bacterial antibiotic resistance, J. Ind. Microbiol. Biotechnol. 29 (2002) 326–330.
- [16] C.M. Magin, S.P. Cooper, A.B. Brennan, Non-toxic antifouling strategies, Mater. Today 13 (2010) 36–44.
- [17] M.P. Schultz, C.J. Kavanagh, G.W. Swain, Hydrodynamic forces on barnacles: implications on detachment from fouling-release surfaces, Biofouling 13 (1999) 323–335.
- [18] R.F. Brady, Fouling-release coatings for warships, Defence Sci. J. 55 (2005) 75–81.
- [19] K. Truby, C. Wood, J. Stein, J. Cella, J. Carpenter, C. Kavanagh, G. Swain, D. Wiebe, D. Lapota, A. Meyer, E. Holm, D. Wendt, C. Smith, J. Montemarano, Evaluation of the performance enhancement of silicone biofouling-release coatings by oil incorporation, Biofouling 15 (2009) 141–150.
- [20] C.J. Kavanagh, G.W. Swain, B.S. Kovach, J. Stein, C. Darkangelo-Wood, K. Truby, E. Holm, J. Montemarano, A. Meyer, D. Wiebe, The effects of silicone fluid additives and silicone elastomer matrices on barnacle adhesion strength, Biofouling 19 (2003) 381–390.
- [21] M. Nendza, Hazard assessment of silicone oils (polydimethylsiloxanes, PDMS) used in antifouling-/foul-release-products in the marine environment, Mar. Pollut. Bull. 54 (2007) 1190–1196.
- [22] F. Azemar, F. Fay, K. Réhel, I. Linossier, Development of hybrid antifouling paints, Prog. Org. Coat. 87 (2015) 10–19.
- [23] M. Rutkowska, M. Jastrzębska, H. Janik, Biodegradation of polycaprolactone in sea water, React. Funct. Polym. 38 (1998) 27–30.
- [24] S.C. Woodward, P.S. Brewer, F. Moatamed, A. Schindler, C.G. Pitt, The intracellular degradation of poly-epsilon-caprolactone, J. Biomed. Mater. Res. 19 (1985) 437–444.
- [25] K.J. Wynne, G.W. Swain, R.B. Fox, S. Bullock, J. Uilk, Two silicone nontoxic fouling release coatings: hydrosilation cured PDMS and CaCO₃ filled, ethoxysiloxane cured RTV11, Biofouling 16 (2000) 277–288.
- [26] D. Wendt, G. Kowalke, J. Kim, I. Singer, Factors that influence elastomeric coating performance: the effect of coating thickness on basal plate morphology, growth and critical removal stress of the barnacle *Balanus amphitrite*, Biofouling 22 (2006) 1–9.
- [27] Y. Tokiwa, B.P. Calabia, Biodegradability and biodegradation of poly(lactide), Appl. Microbiol. Biotechnol. 72 (2006) 244–251.
- [28] F.D. Kopinke, M. Remmler, K. Mackenzie, M. Möder, O. Wachsen, Thermal Decomposition of Biodegradable Polyester – II. Poly(Lactic Acid), (2020).
- [29] A. Kamsikakis, E. Kavetsou, K. Chronaki, E. Kiosidou, E. Pavlatou, A. Karana, C. Papaspyrides, A. Detsi, A. Karantonis, S. Vouyiouka, Encapsulation of antifouling organic biocides in poly(lactic acid) nanoparticles, Bioengineering 4 (2017) 81–99.
- [30] M. Thouvenin, V. Langlois, R. Briandet, J.Y. Langlois, P.H. Guerin, J.J. Peron, D. Haras, K. Vallee-Rehel, Study of erodable paint properties involved in antifouling activity, Biofouling 19 (2003) 177–186.
- [31] P. Shen, A. Moriya, S. Rajabzadeh, T. Maruyama, H. Matsuyama, Improvement of the antifouling properties of poly (lactic acid) hollow fiber membranes with poly (lactic acid)-polyethylene glycol-poly (lactic acid) copolymers, Desalination 325 (2013) 37–39.
- [32] F. Azemar, F. Fay, K. Réhel, I. Linossier, Control of hydration and degradation properties of triblock copolymers polycaprolactone-b-polydimethylsiloxane-b-polycaprolactone, J. Appl. Polym. Sci. 12 (2014) 131–138.
- [33] E. Almeida, T.C. Diamantino, O. De Sousa, Marine paints: the particular case of antifouling paints, Prog. Org. Coat. 59 (2007) 2–20.
- [34] M. Lorient, I. Linossier, K. Vallée-Réhel, F. Fay, Influence of biodegradable polymer properties on antifouling paints activity, Polymers 9 (2017) 36.
- [35] Y. Baek, J. Kang, P. Theato, J. Yoon, Measuring hydrophilicity of RO membranes by contact angles via sessile drop and captive bubble method: a comparative study, Desalination 303 (2012) 23–28.
- [36] D. Garlotta, A literature review of poly(lactic acid), J. Polym. Environ. 9 (2001) 63–84.
- [37] F. Fay, I. Linossier, J.J. Peron, V. Langlois, K. Vallée-Réhel, Antifouling activity of marine paints: study of erosion, Prog. Org. Coat. 60 (2007) 194–206.
- [38] ISO 15181-1, Paints and Varnishes – Determination of Release Rate of Biocides from Antifouling Paints - Part 1: General Method for Extraction of Biocides, (2007) (review 2017).
- [39] ISO 15181-15182, Paints and Varnishes - Determination of Release Rate of Biocides from Antifouling Paints - Part 2: Determination of Copper-ion Concentration in the Extract and Calculation of the Release Rate, (2007) (review 2017).
- [40] A.S. Adeleye, E.A. Oranu, M. Tao, A.A. Keller, Release and detection of nanosized copper from a commercial paint, Water Res. 102 (2016) 374–382.
- [41] M. Lagerström, J.F. Lindgren, A. Holmqvist, M. Dahlström, E. Ytreberg, *In situ* release rates of Cu and Zn from commercial antifouling paints at different salinities, Mar. Pollut. Bull. 127 (2018) 289–296.
- [42] J.F. Briand, A. Barani, C. Garnier, K. Réhel, F. Urvois, C. LePoupon, A. Bouchez, D. Debroas, C. Bressy, Spatio-temporal variations of marine biofilm communities colonizing artificial substrata including antifouling coatings in contrasted French coastal environments, Microb. Ecol. 74 (2017) 585–598.
- [43] D. Haras, Biofilms and deteriorations of materials: from analysis of the phenomenon to prevention strategies, Mater. Technol. 93 (2006) 27–41.
- [44] NF T 34-552, Peintures et vernis – Systèmes de peintures pour la protection des ouvrages en acier – Essai d’immersion au radeau en eau de mer vive, (1996).